Question 11: Installing Thrift, generating HBase Thrift binding and interacting with HBase

interacting with HBase

Aim

To install Apache Thrift, generate client bindings for the HBase Thrift interface and use a client to interact with HBase.

Procedure

1. Install a Thrift compiler version compatible with the HBase Thrift interface and install the required language runtime,

such as Python.

2. Download the HBase source or distribution containing the Thrift interface definition file (Hbase.thrift).

3. Start HBase and start the HBase Thrift service using the script provided by the installed HBase release.

4. Generate client-side bindings from Hbase.thrift using the Thrift compiler.

5. Configure the generated client with the Thrift server hostname and port.

6. Use the generated client to connect to HBase and perform table operations. The exact generated API names can

differ between Thrift versions.

Commands

Install Thrift using the package manager or the official

Apache Thrift installation instructions.

thrift --version

Locate Hbase.thrift in the HBase distribution/source tree.

Generate Python bindings (output folder is created if needed):

mkdir -p gen-py

thrift --gen py -out gen-py Hbase.thrift

Start HBase, then start the Thrift service.

start-hbase.sh

hbase thrift start

The default Thrift port is commonly 9090.

Example Python client (legacy HBase Thrift API)

from thrift.transport import TSocket, TTransport

from thrift.protocol import TBinaryProtocol

```
from hbase import Hbase

socket = TSocket.TSocket('localhost', 9090)

transport = TTransport.TBufferedTransport(socket)

protocol = TBinaryProtocol.TBinaryProtocol(transport)

client = Hbase.Client(protocol)

transport.open()

print(client.getTableNames())

transport.close()
```

Output of Program

```
Connected to HBase Thrift server.

Tables:

['student']
```

Result

Thrift bindings were generated and the client connected to the HBase Thrift service to retrieve the table list.

Important: HBase Thrift APIs have changed across releases. Use the Hbase.thrift file shipped with the exact installed HBase

version and its matching generated client. Some releases use a separate hbase thrift start command or service script.